GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY

Block B, 7th Floor, M. S. Building, Nr. Pathikashram, Sector 11, Gandhinagar-382011

Robofest Application

Application Number :- R520250863

Date : 25-Jul-2025

Mentor Name :- Maryam Kaveshgar

Mentor Email Id:- thecafestory15@gmail.com

Institution Name :- Ahmedabad University

Complete Postal Address :- Ahmedabad University, Commerce Six Roads, Navrangpura, Ahmedabad 380009, Gujarat, India

Name of Affiliated University/Board Name :- Ahmedabad University

Experience Of Mentor/Coach/Faculty In Years :- 10Y

Robot Making Capability :- The School of Engineering and Applied Science at Ahmedabad University is equipped with a Mobile Robots Lab. The Mentor and students have already presented their work and participated in national and international events. Students have access to the state-of-the-art Tinkerers Lab.

Mentor Expertise :- autonomous mobile robots, navigation and commonication, ITS

Mentor Mobile Number :- 8487014815

Type of Institution :- Private-Unaided

Proposal Submitted For Robot :- Autonomous Underwater Vehicle

Pin Code :- 380009

Office No :-

Participation List :- yes,BFMC(Bosch Future Mobility challenge), robofest

Proposed Team of Making robot :

Name Of Student/Research Scholar	Institute Name	Department Name	Programme Enrolled	Year	Mobile No
Foram Gohel	Ahmedabad University	Fifth Semster	BTECH CSE	2023	8487014815
Kahan Bhatt	Ahmedabad University	Fifth Semster	BTECH MECHANICAL	2023	9427424409

Dev Kansara	Ahmedabad University	Fifth Semster	BTECH CSE	2023	7990943632
Yash Shah	Ahmedabad University	Fifth Semester	BTECH CSE	2023	8849635872

GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY
Dept. of Science and Technology, Govt. of Gujarat

Application format for the
IDEATION STAGE FOR ROBOT MAKING COMPETITION
ROBOFEST GUJARAT-5.0

١. Proposal submitted for Robot: (Please tick mark the category of robot proposed)

- **For Junior Level:**
  1. AI Homo-bots
  2. AI Autonomous Driving Bots
  3. Maze Solving Robot
  4. Robot Athletics

- **For Senior Level:**
  - ☑ AerialRobotics: Minefield Navigation Challenge
    - ○ Autonomous Underwater Vehicle
    - ○ Autonomous Maze solver
    - ○ Intelligent Ground Vehicle Competition
    - ○ Application Based Categories
      - ■ Robots for healthcare
      - ■ Robots for humanitarian propose
      - ■ Robot for educational/research purposes
      - ■ Robots for community services or community awareness
      - ■ Robots for tackling the climate change challenges or assisting with climate
      - ■ sustainability
      - ■ Robots for industrial applications and
      - ■ Robots for world peace, for example, surveillance, narcotics control etc.
  -

٢. Type of Institution:

☐ Government: □ Private-Aided: □ Private-Unaided: ✓ Local Body: □

٣. Name of School/Institute/Department/Organization: Ahmedabad University

٤. Complete Postal Address with Pin code: Ahmedabad University, Commerce Six Roads, Navrangpura, Ahmedabad 380009, Gujarat, India

٥. Name of affiliated University/Board Name:

٦. Mentor/Coach/Faculty Guide Name: Maryam Kaveshgar

٧. Designation of Mentor/Coach/Faculty Guide: Assistant Professor

٨. Experience of Mentor/Coach/Faculty in years: 10 Y

٩. Email address of mentor: maryam.kaveshgar@ahduni.edu.in

١٠. Mobile Contact No of mentor: 9726513322

١١. Office No: 79-61911144

١٢. Proposed Team of Making robot: (Maximum four participant per team allowed)

Sr. No.	Name of Student/Research scholar	Institute Name	Department	Programme enrolled	Standard / Year	Mobile No.	E-mail address
1	Foram Gohel	Ahmedabad University	Computer Science	B.Tech	3	8487014815	foram.g@ahduni.edu.in
2	Kahan Bhatt	Ahmedabad University	Mechanical	B.Tech	3	9427424409	kahan.b@ahduni.edu.in
3	Dev Kansara	Ahmedabad University	Computer Science	B.Tech	3	7990943632	dev.k1@ahduni.edu.in
4	Yash Shah	Ahmedabad University	Computer Science	B.Tech	3	8849635872	yash.s14@ahduni.edu.in

١٣. Capability of the Organization / Individual specific to Robot Making:

The School of Engineering and Applied Science at Ahmedabad University is equipped with a Mobile Robots Lab. The Mentor and students have already presented their work and participated in national and international events. Students have access to the state-of-the-art Tinkerers Lab.

a. Expertise available with the Mentor/Coach/Faculty Guide: (Not more than 100 words)
Mentor has extensive experience in developing autonomous mobile robots from small-scale to autonomous vehicles. Her field of research is focused on control systems and automation, with a special interest in Intelligent Transportation. She has guided and mentored multiple teams in national and international events.
She had mentored student teams in previous ROBOFEST events and has secured awards. She has multiple filed patents and publications in reputed journals and proceedings. She is the BoG member of IEEE ITSS.

b. List of Participation in past in any Robot Making Competitions:

Event Name	Venue	Type of Competition (State/national/International)	Type of Robot submitted	Achievement
BFMC(Bosch Future Mobility challenge)	Romania	International	Small scale autonomous vehicle	Award
Robofest 3.0	Gujarat	National	Rover	Award, Winner
Robofest 4.0	Gujarat	National	Others	Ideation phase clearence
BFMC(Bosch Future Mobility challenge)	Romania	International	Small scale autonomous vehicle	Award

Ahmedabad University

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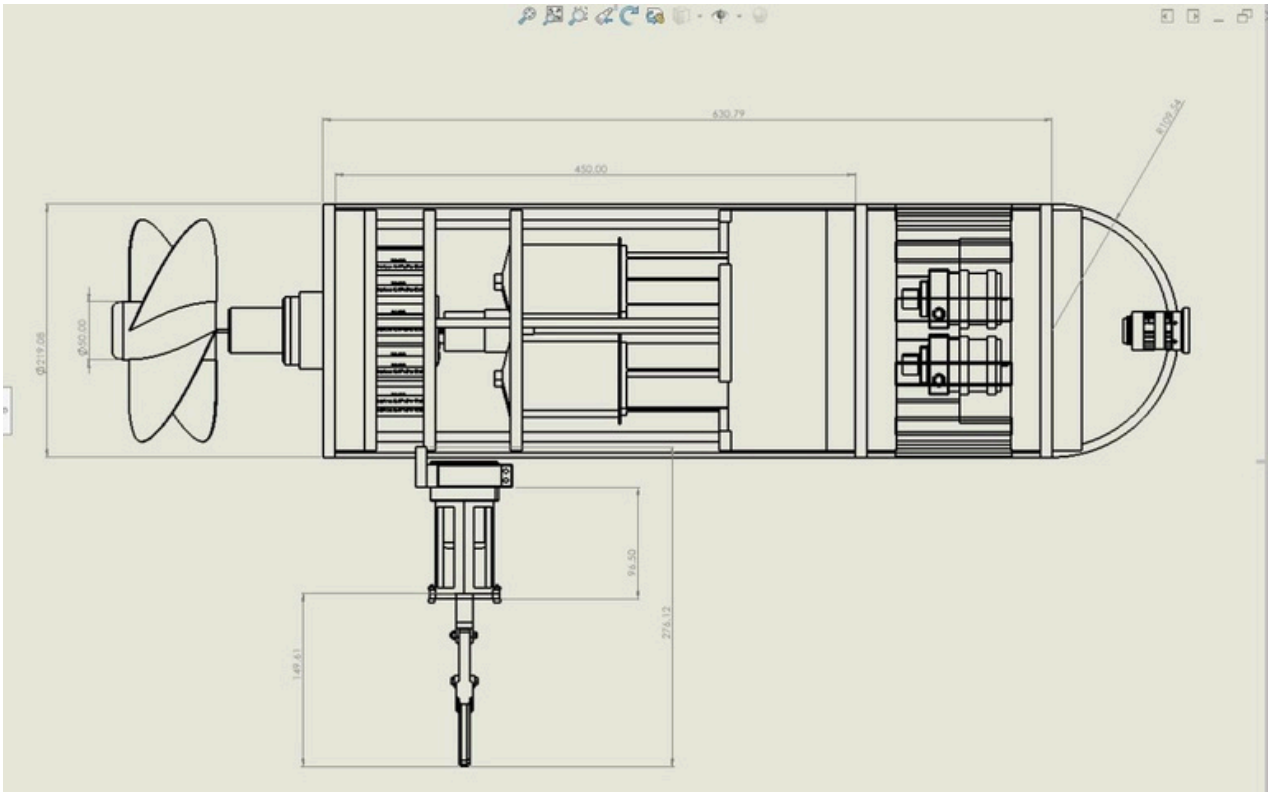
SCHOOL OF ENGINEERING AND APPLIED SCIENCE * AHMEDABAD UNIVERSITY

TECHNICAL DETAILS FOR PROPOSED ROBOT

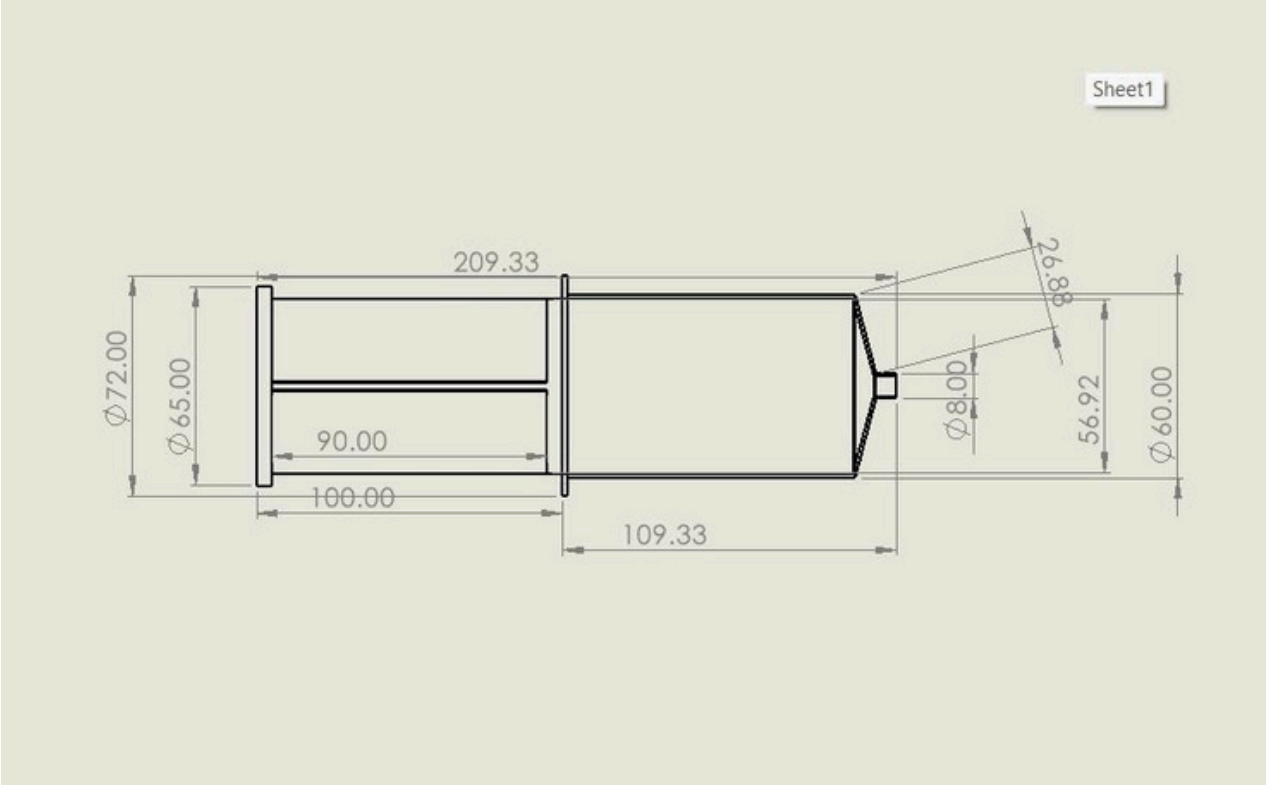
1. Type of Robot: Autonomous Underwater Vehicle

2. Robot Assembly Design (Proposed Diagram): Drawings for each part of the robot are preferred as an attachment. (CAD drawings are preferred)

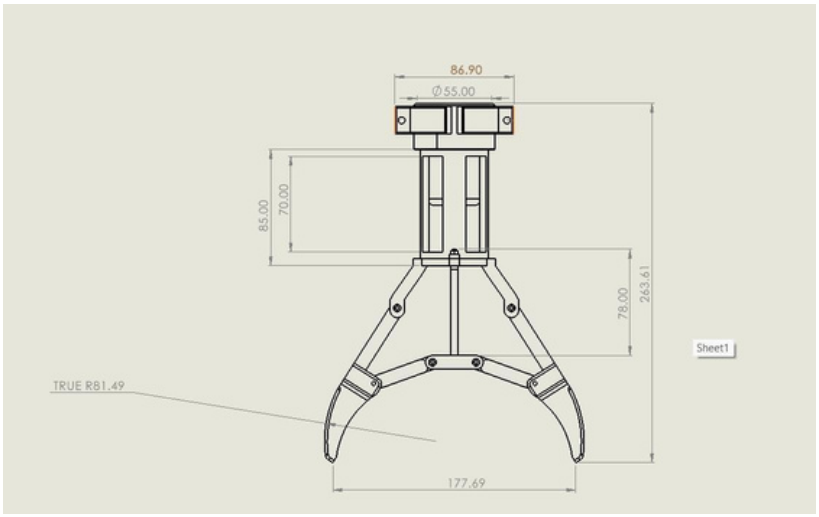
Final Assembly with basic Dimensions



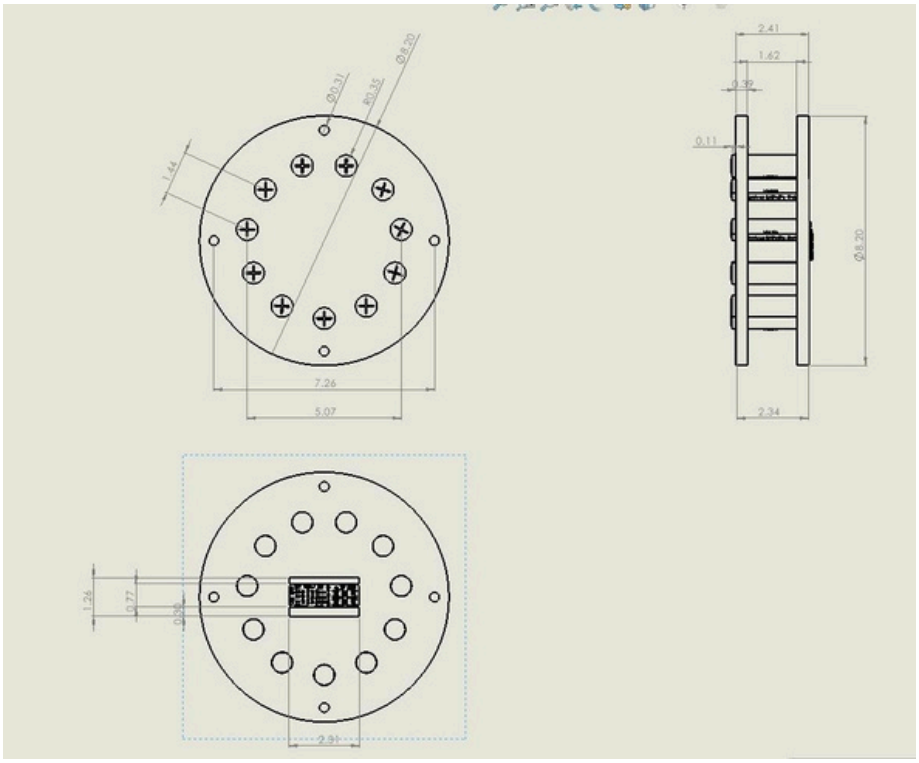
Single syringe (use for ballast tanks)



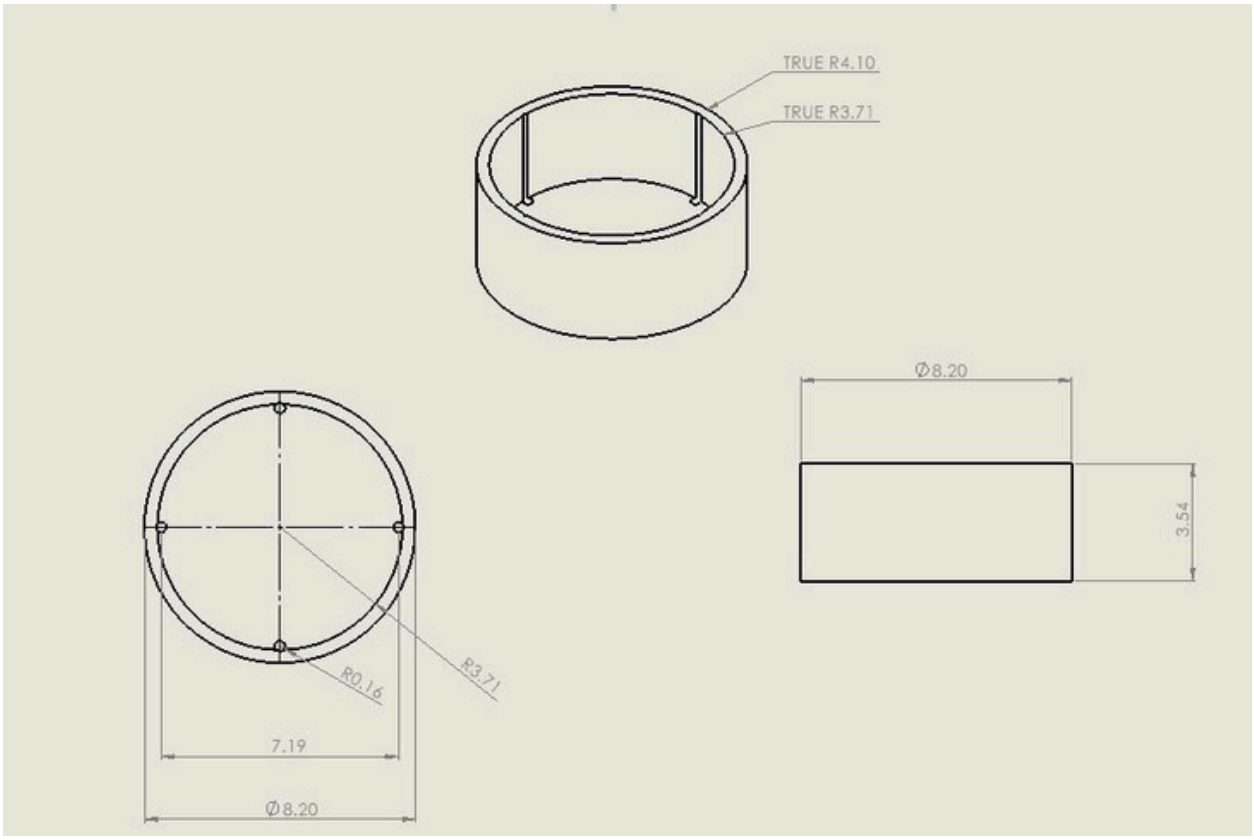
Gripper



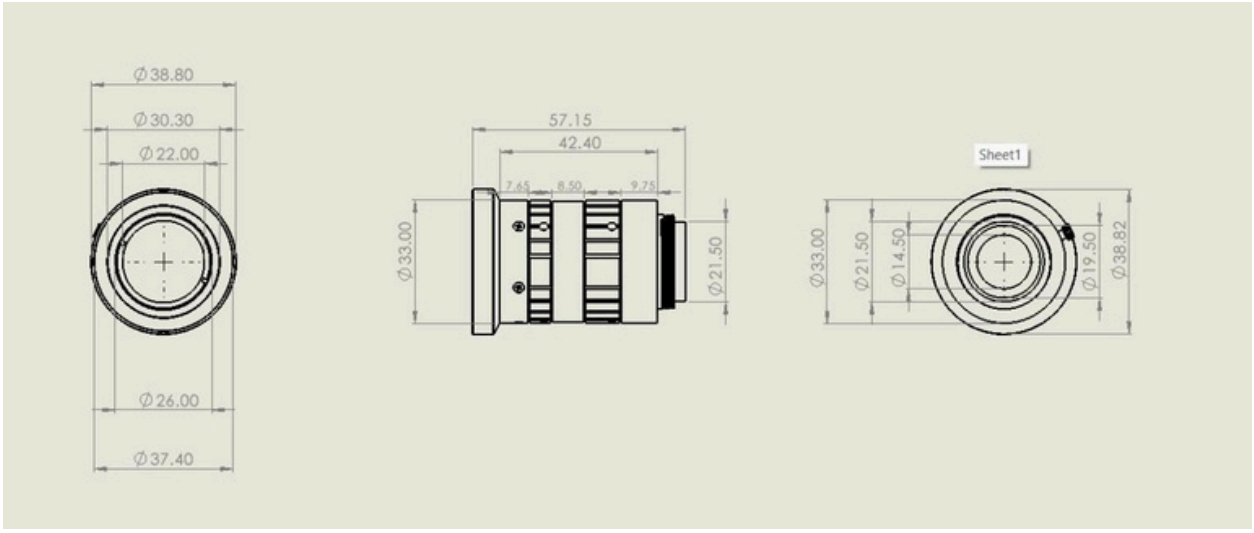
Lithium Thingies



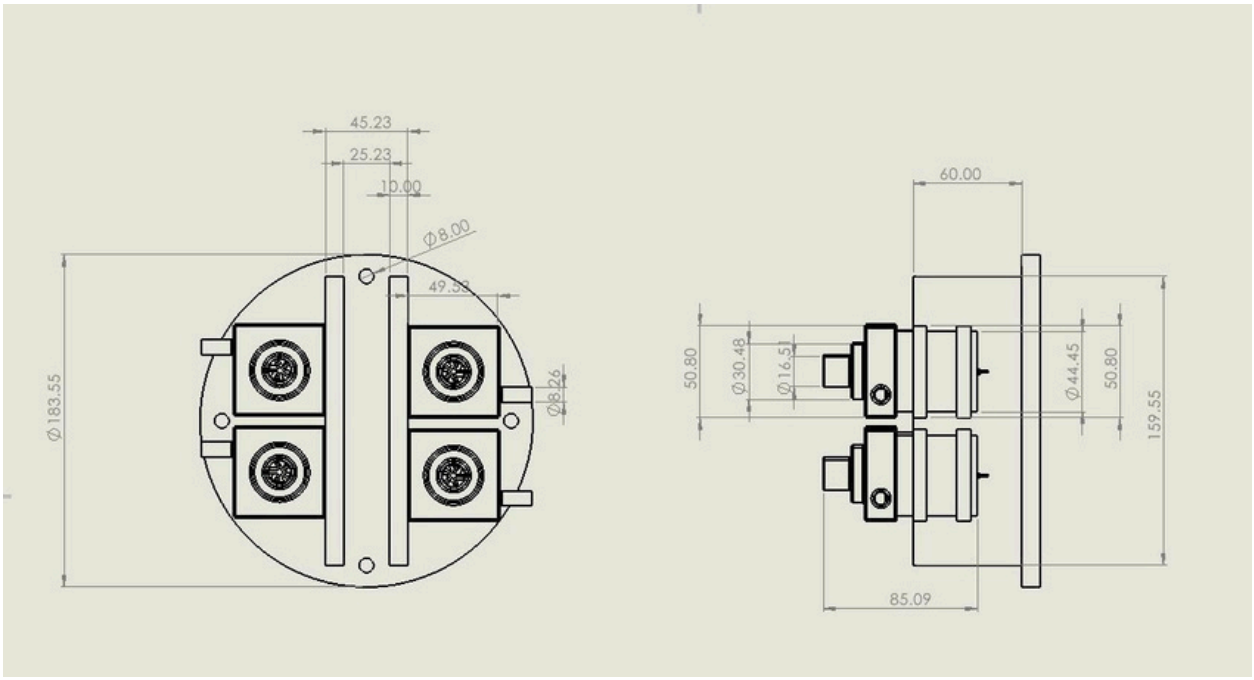
sensor box



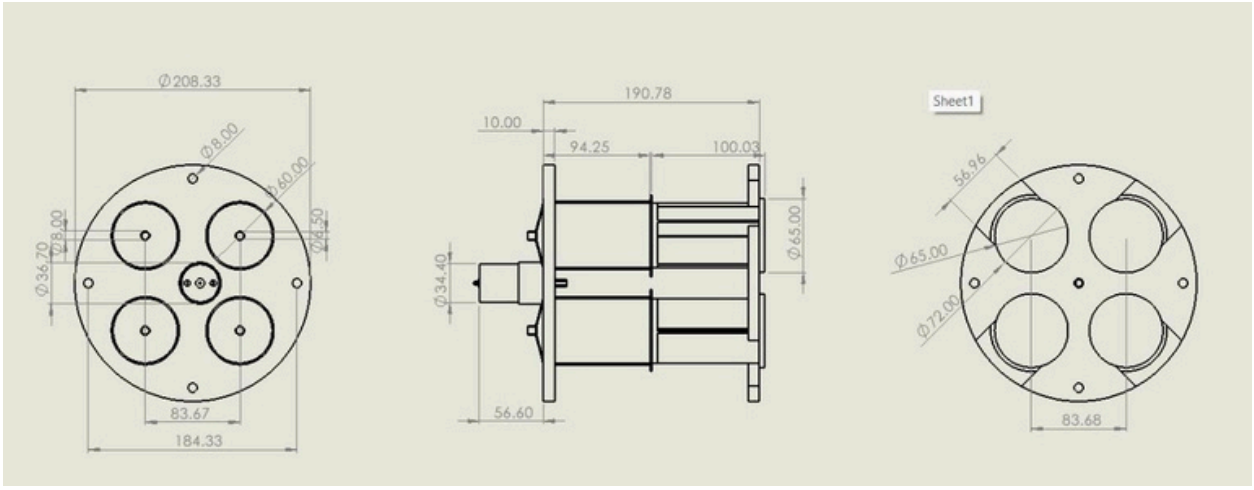
Camera



Turn pumps(used as jets)



Ballast Tank



1. Components to be used: (Enlist all the components with their make/company in four groups as enlisted in the following table:
   a. List of Structure components: like beams, bushes, shafts, belts, plates, pins, pulleys, wheels,

      connectors, batteries, motors, etc.

Component	Material / Example	Purpose	Notes
Front Dome	Acrylic / Polycarbonate	Optical window for front camera	Transparent, optical quality
End Caps	Aluminum, anodized,	Sealing ends of hull	Removable for

(bulkheads)	O-ring sealed		access
Internal Rails/Beams	Aluminum extrusion (2020) / flat bars	Mounting electronics & batteries	Structural support
Mounting Plates	Aluminum / ABS / 3D printed	Holding cameras, pumps, pistons, gripper	Custom cut or CNC
Brackets	Aluminum / Stainless steel	Fix thruster, pumps, gripper, payload	Reinforced
Shafts	Stainless steel rods (Ø6–10 mm)	Pivots, feedthrough rod for gripper	Corrosion resistant
Bushes / Bearings	Nylon/bronze bushings, sealed bearings	Smooth motion of pivots & jaws	Waterproof grease
Pins & Fasteners	SS dowel pins, cotter pins, screws, circlips	Assembly & linkages	Marine-grade
Springs	Stainless steel compression/extension springs	Fail-safe jaw open bias (gripper)	Optional
Thruster Mount	Alu/SS plate with Kort nozzle	Holds rear thruster	Strength critical
Pump Mounts	Side plates with jet ducts	Fix water pumps for yaw/sway	2 left, 2 right
Piston Mounts	Square pattern brackets	Hold ballast pistons around CG	4 pistons total
Payload Dropper	Servo bracket + gate plate	Drop 70 mm ball	Simple hinge/servo
Gripper Jaws (2)	POM/Nylon plates with rubber pads	Grip payloads	Parallel 2-jaw
Gripper Linkages	Aluminum 4-bar + crossbar	Transmit actuator motion to jaws	External
Dynamic Rod Feedthrough	SS shaft + Viton O-rings	Pass linear motion out of hull	Sealed
Connectors (Waterproof)	IP68 glands, SubConn connectors	Cable feedthroughs	Critical sealing
Battery Housing	Internal tray with straps/rails	Secure 6S Li-ion battery	Shock resistant
Motor Mounts	Plates for ESCs, linear actuators	Secure actuators in hull	With cooling space
Vacuum Test Port	Valve + gauge	Hull leak testing	Mandatory before dives
Desiccant Holders	Small trays inside hull	Moisture absorption	Replace regularly

b. List of Motion Components: like Chain, sprockets, flaps, etc.

Component	Type / Example	Purpose (Motion Axis)	Notes
Rear Thruster	BLDC thruster + propeller	Surge (forward/backward)	Provides main propulsion
Water Pumps (x4)	Centrifugal/diaphragm pumps with nozzles	Yaw & minor sway	Two per side for torque/translation
Pistons (x4)	Syringe-style linear actuators	Heave (up/down)	Adjust buoyancy/displacement
Gripper Feedthrough Rod	Stainless steel sliding rod (sealed)	Transfers actuator motion	From inside hull to outside jaws
Payload Dropper Servo	Micro/standard waterproof servo + flap gate	Release payload ball	Rotates flap to drop 70 mm sphere
Flaps/Nozzles (pump ducts)	Jet outlets / ducts	Direct pump flow	Guide yaw/sway jets efficiently
Shafts (pivot rods)	Stainless steel rods	Pivot for jaws & linkages	With bushings for smooth motion
Springs (optional)	SS coil springs	Fail-safe bias for jaws	Default open on power loss

a. List of Electronics Components: like Smart ports, switches, joysticks, controllers, LED/LCD screen, power supply, programming components, etc.

Category	Component	Example / Model	Purpose
Computing & Control	Single Board Computer (SBC)	Jetson Orin Nano / Raspberry Pi 5	High-level mission control, mapping, vision
	Microcontroller (MCU)	STM32F4/F7 / Teensy 4.x / Arduino Mega	Real-time motor, pump, piston, gripper control; PID loops
	Motor Controllers (ESCs)	BLDC ESCs (60A thruster, 10–20A pumps)	Drive thruster & pumps
	Stepper / Servo Drivers	Stepper drivers,	Drive pistons & gripper

		waterproof servo driver	actuator
	Power Distribution Board (PDB)	Custom PCB with fuses & filters	Safe power distribution
	DC-DC Converters	22V→12V, 22V→5V regulators	Power electronics & sensors
	Relays / Solid-State Switches	60A relay / MOSFET switch	Cut-off for thruster/pumps
	Programming Interfaces	USB-C, UART, SWD/JTAG	Firmware flashing, debugging
Power System	Battery Pack	6S Li-ion, 22.2V, 15–20Ah	Main power source
	Battery Management System (BMS)	6S, 60A continuous	Protects cells (OV, UV, OC)
	Magnetic Reed Switch	Waterproof kill switch	Safety arming/disarming
Vision Sensors	Front Camera	USB3 1080p60 global shutter	Gate detection & navigation
	Downward Camera	USB3 1080p60 wide-angle	Mat detection, drum ID, gripper alignment
	LED Ring / Lights	Waterproof LED ring	Lighting for underwater vision
Navigation Sensors	IMU	9-axis IMU (Accel + Gyro + Mag)	Attitude, yaw, stabilization
	Pressure/Depth Sensor	0–3 bar waterproof sensor	Depth measurement & control
	Hydrophones (x3) + Preamps	Omnidirectional hydrophones (1–50 kHz)	Detect pinger direction
	Hydrophone ADC	24-bit, 192 kHz synchronized ADC	Acoustic signal acquisition
	Compass (backup)	HMC5883L or integrated in IMU	Heading estimation
	GPS (surface only, optional)	Waterproof GPS receiver	Surface localization (not underwater)
Mapping System	Depth + IMU Fusion	IMU + Pressure sensor	Position in Z-axis & orientation
	Dead-reckoning System	Encoder feedback + IMU	Short-term position tracking
	Vision-based Mapping	Front + Down cameras (ORB-SLAM/RTAB-Map)	Underwater SLAM for object localization
	Acoustic Positioning	Hydrophones + ADC	Triangulation with pinger

	(optional)		
Manipulation Sensors	Load Cell (Gripper)	50–200N S-beam + HX711 amplifier	Grip force measurement
	Limit Switches / Reed Sensors	Waterproof microswitches (2–4 pcs)	Detect jaw open/close
	Encoder (Actuator/Motor)	Integrated in stepper/linear actuator	Position feedback
Safety Sensors	Leak Detector	Optical water / humidity sensor	Leak detection inside hull
	Internal Temperature Sensor	DS18B20 / Thermistor	Monitor electronics & battery temperature
	Voltage & Current Sensor	INA219 / Hall-effect sensor	Battery voltage & current monitoring
	Internal Humidity Sensor	DHT11/DHT22	Early leak/moisture detection
Data & Storage	SSD / SD Card	NVMe SSD (128–256 GB) / SDHC	Store mission logs, video, sensor data

b. List of Other Accessories: Clothes, plastic eyes/ear/feeling like real all external components which are for the look.
No extra addition for looks

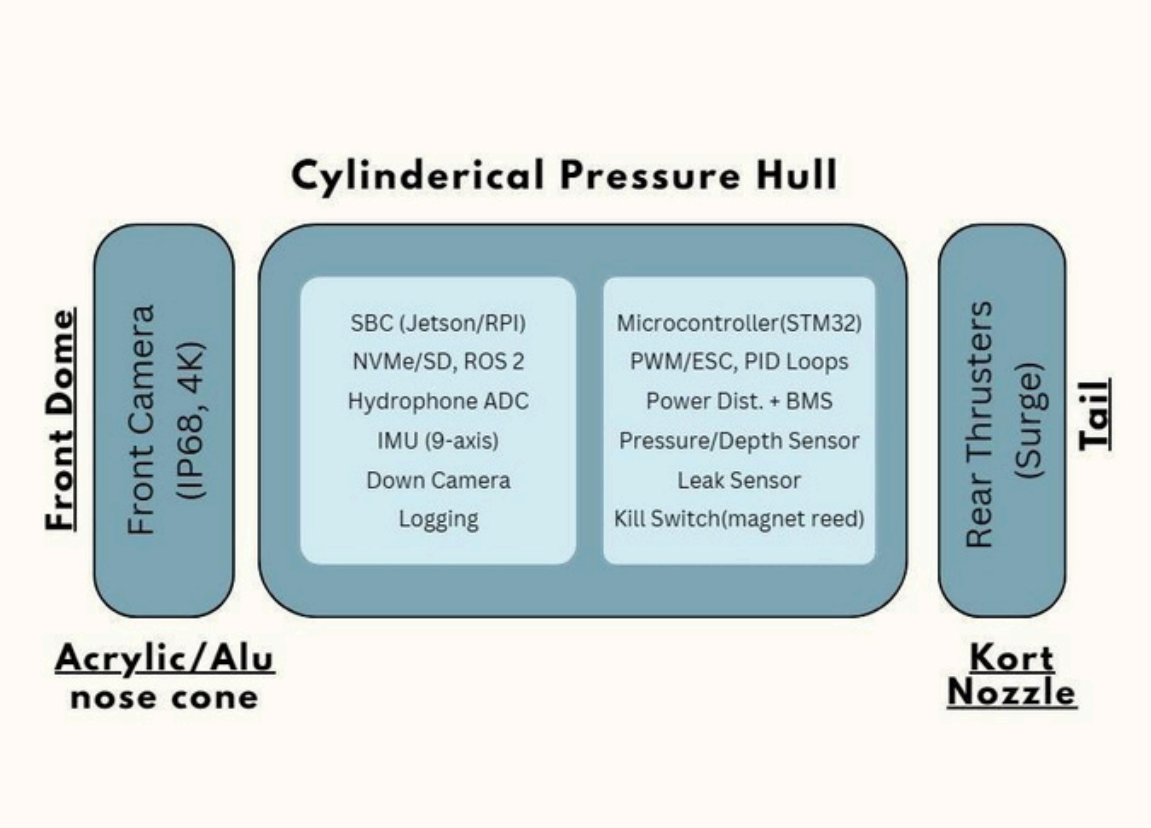
2. Themethodology of Making Robot:(Please write Entire Technical Specifications of Proposed Robot with brief notes and diagrams)

Main Concept

- **Form factor:** sealed cylindrical hull (submarine-style), neutrally buoyant, with removable nose &tail caps.
- **Actuation concept:**
  - 1rear thruster for *(surge:forward/back)*.
  - 4water pumps work as **lateraljets** *(2left, two right)* for **yaw + minor sway** control.
  - 4mini **pistons (syringe-style ballast)** for **heave (up/down)** via variable displacement (closed-loop buoyancy control).

- **Sensing:** front camera (targeting/gate), downward camera (mat & drums), 3-hydrophone “Y” array for pinger direction, IMU,depth/pressure sensor, leak sensor.
- **Compute:** onboard SBC (e.g.,Jetson Orin Nano / RPi 5) + microcontroller (STM32/Teensy) for hard-real-time motor loops.
- **Mission:** fully autonomous—nosurfacing, no wall/floor contact; do gate, mat approach, drum ID (blue and pinger), ball drop,optional 2D map, return & surface.

system architecture (block)



Mechanical Design

Body Design

Geometry: Body size will be around 160-200 mm in diameter, and the length of the cylinder will be around 650-800 mm *(choose to fit buoyancy calcs)*

Material: PVC pipe for body and acrylic for partitions

Seals: dual radial O-rings on nose & tail bulkheads; cable penetrators (Subconn/BlueTrail kind) for pumps/thrusters; desiccant pack inside.

Mounting rails: internal 2020 profiles for electronics trays and piston manifold.

Depth rating: target 5–10 m (more than the 1.5 m+ pool), FOS ≥ 2.

- **Actuators**

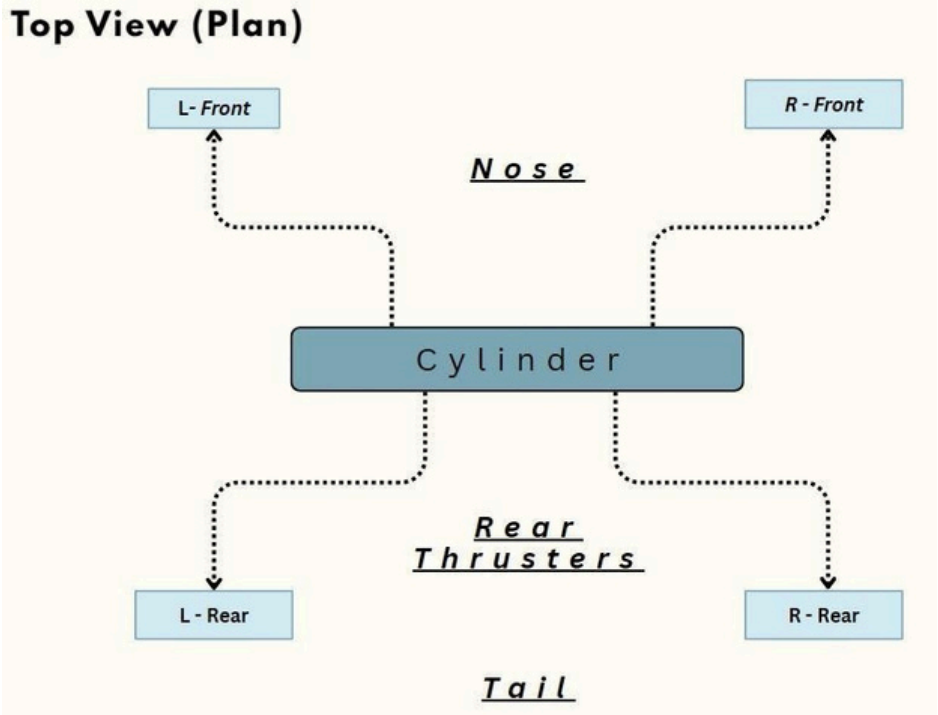
  Rear thruster -

  Specifications - 200–300 W brushless thruster, bollard thrust ~35–50 N; ESC with reverse.

  Function - surge (forward/back) + coarse speed control for approach/return.

  4 water pumps (lateral jets)

- **type:** brushlessDCdiaphragmorcentrifugal pumps (IP68 or inboard with sealed jet nozzles).
- **Layout:** twonozzlesperside(fore&aft) pointing **outboard**.
- **Control of all types of movements:**
  - **Yaw:** LeftPair↑ RightPair↓ (differential thrust).
  - **Sway (minor):** alllefton to move right (reaction jet), all right on to move left.
  - Use PWM for proportional thrust steps.



  4 pistons (heave)

- **Concept:** syringe-style linear actuators (e.g., 60–100 ml each) moving water in/out of internal bladders to changedisplacement without opening hull to pool water (safer).

- **Arrangement:** Use a square pattern around CG for trim; use two for coarse buoyancy and two for trim pitch/roll.
- **Stroke/Time:** full stroke < 3 s recommended for responsive depth hold near drums.
- **Feedback:** linear potentiometer (or step count) per piston.

buoyancy & stability (quick design math)

- **Goal:** slightly positively buoyant at rest, pistons pull in to sink and push out to rise.

stability: keep heavy battery and thruster low; pistons symmetrically placed to minimize trim change.

electrical & power

- **Battery:** 6-cell Li-ion (22.2 V) 10–20 Ah; **BMS** with 60–80 A burst; waterproof quick-disconnect.
- **PDN:** 22 V bus → 12 V (pumps) & 5 V (SBC, cameras, hydro ADC) via isolated DC-DCs; fusing per branch; TVS diodes; common ground star point.
- **Connectivity:** internal CAN (MCU network) + USB 3 (cameras) + I²C/SPI (IMU, pressure).
- **Kill & safety:** magnetic reed external kill; internal leak detector (optical).
- **Runtime:** >10 min continuous thrust per guidelines

Sensing & perception

Cameras (vision-only navigation)

- **front cam:** At the acrylic dome port to avoid refraction distortion.
- **Down cam:** angled slightly forward to see drums on the mat.

Vision tasks & cues (all color/shape robust to pool lighting):

- **Gaterun:** detect rectangular orange/black gate rails (or provided color); use Hough lines + contour geometry; maintain centerline & pass through; 180° U-turn; reverse pass. Scored if no

wall/floor contact.
ViewFile

- **Green mat approach:** color threshold in HSV for green range; downward cam for mat ROI; PID to center over mat's centroid.
ViewFile

- **Drumidentification:** segment circular drums; classify **blue vs red** via HSV/area; resolve **one blueamong threered** .

Passive acoustics

- **array:** 3 hydrophones (triangle/Y-shape, 120–200 mm spacing) on underside.

- **Pipeline:** band-pass around pinger frequency (given by organizers), cross-correlate channels for TDoA → bearing estimate; fuse with IMU for yaw command toward **pinger drum**. (This remains "passive"—you are not emitting sound.) The rules expect pinger localization ability.

Inertials & depth

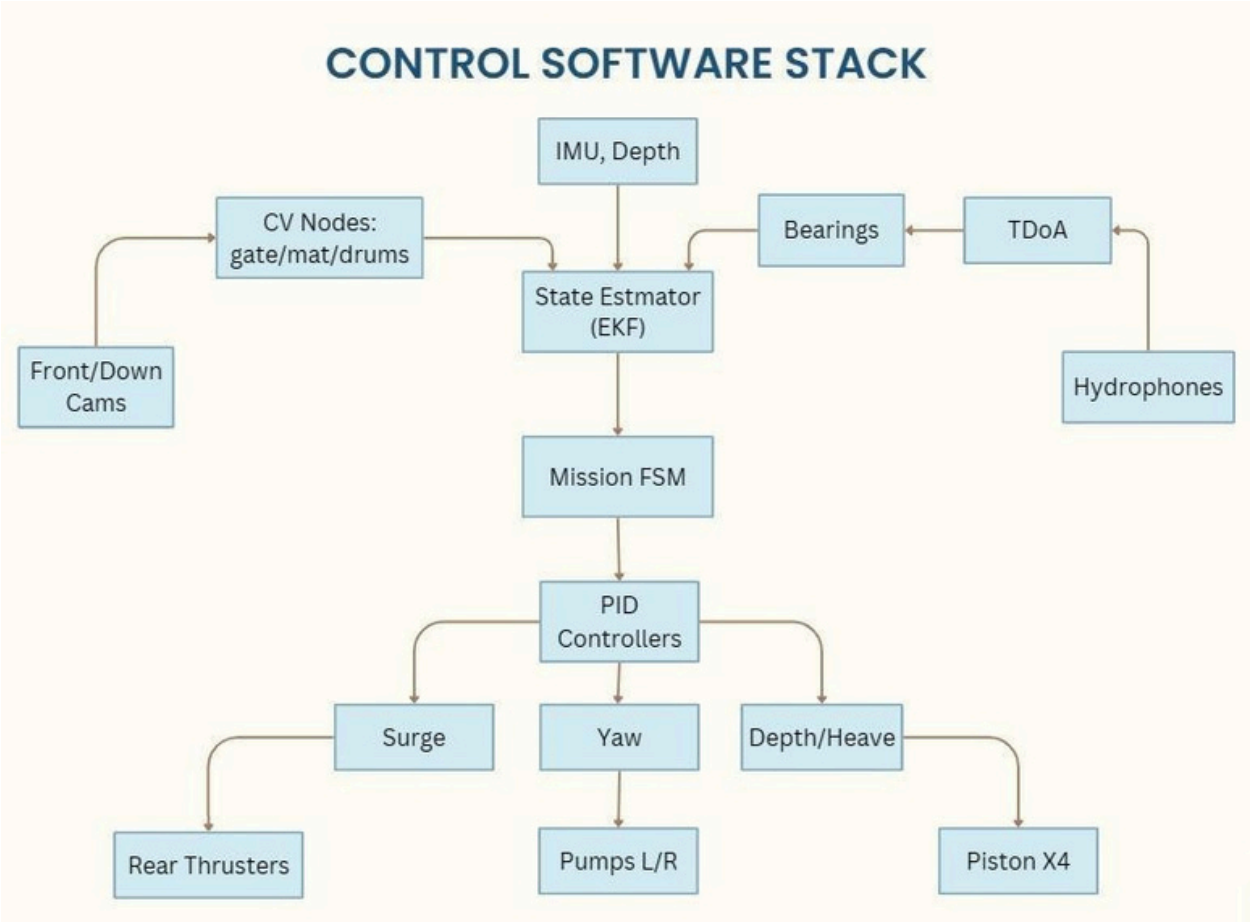
- **IMU:** 9-axis with factory calibration; Mahony/Madgwick on MCU

- **pressure sensor:** 0–3 bar absolute; depth loop

- **Odometry:** dead-reckon from IMU + surge velocity (from ESC RPM model) for short missions; vision beacons (mat centroid) for drift reset.

Software stack

- **nodes:**
    - camera_front_node, camera_down_node
    - vision_gate, vision_mat, vision_drum_classifier
    - hydro_bearing_estimator
    - state_estimator (EKF: IMU+depth+vision bearings)
    - controller (PID: depth, yaw, surge, lateral)
    - actuator_bridge (to MCU over CAN/serial)
    - mission_manager (state machine)
    - mapper_2d (bonus: build 2D schematic of mat + drum positions for post-mission)
    ViewFile
    - logger



control loops

- **Depth PID:** setpointfrommission or when stabilizing at mat height.
- **Yaw PID:** targetfromhydrophone bearing or vision (gate normal, drum azimuth).
- **Surge feed-forward + PI:** for approach/return speed.
- **Lateral control:** pumps driven by yaw error derivative (quick heading snaps) and minor sway correctionsfrom vision centroid offset.

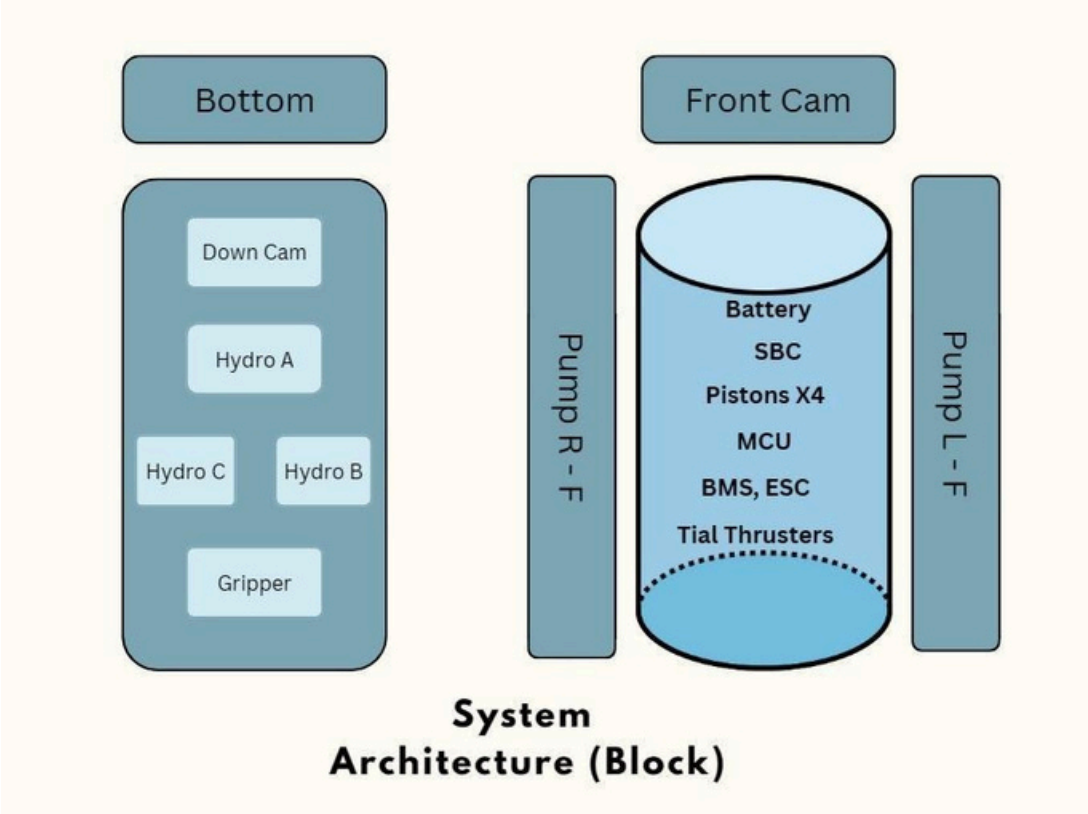
Payload release mechanism

Type:Parallel-jaw,2-finger gripper mounted under hull (outside body).

Actuation:Singleinternal linear actuator (leadscrew/servo) pushes a sealed rod through a feedthrough.Outside,rod drives a crossbar + 4-bar linkage → both jaws move symmetrically.
Specs:

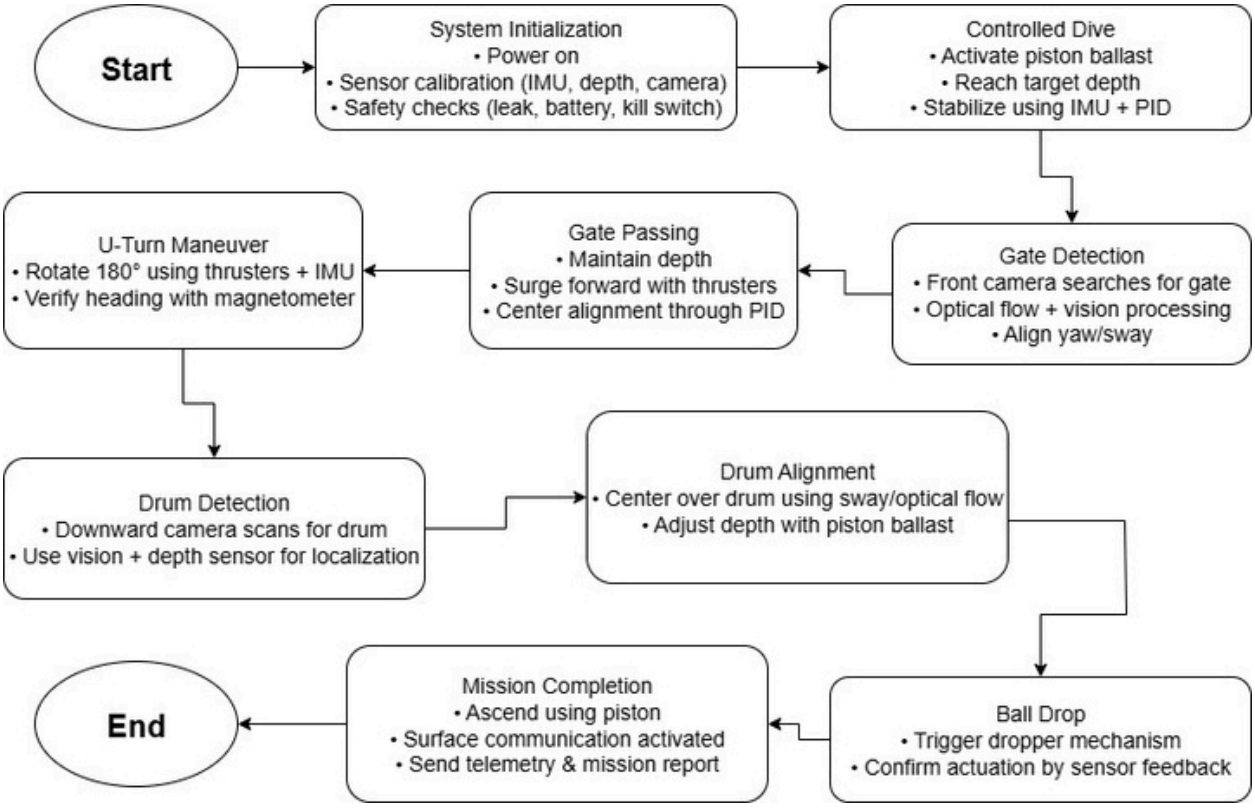
- Jaw opening: 0–120 mm (fits 70 mm ball).

- Grip force: ≥25 N per jaw (design for >100 N axial force).



System Architecture (Block)

Flow of Mission



3. Application of proposed Robot in a societal context: (Not more than 100 words)

The proposed autonomous underwater robot can support critical tasks such as underwater inspection, search and rescue, and marine research. It can navigate confined spaces, pass through gates, and precisely deploy objects, making it suitable for pipeline monitoring, port security, or environmental sampling. By reducing human risk in hazardous underwater missions and providing reliable data collection, the robot contributes to safer operations, improved resource management, and better protection of aquatic ecosystems. Size of robot proposed for Proof of concept (Small version):

4.

a. Length in cm- 40cm b. Width in cm - 15cm c. Height in cm - 15cm

5. Size of robot proposed as Proto type (Actual Version)

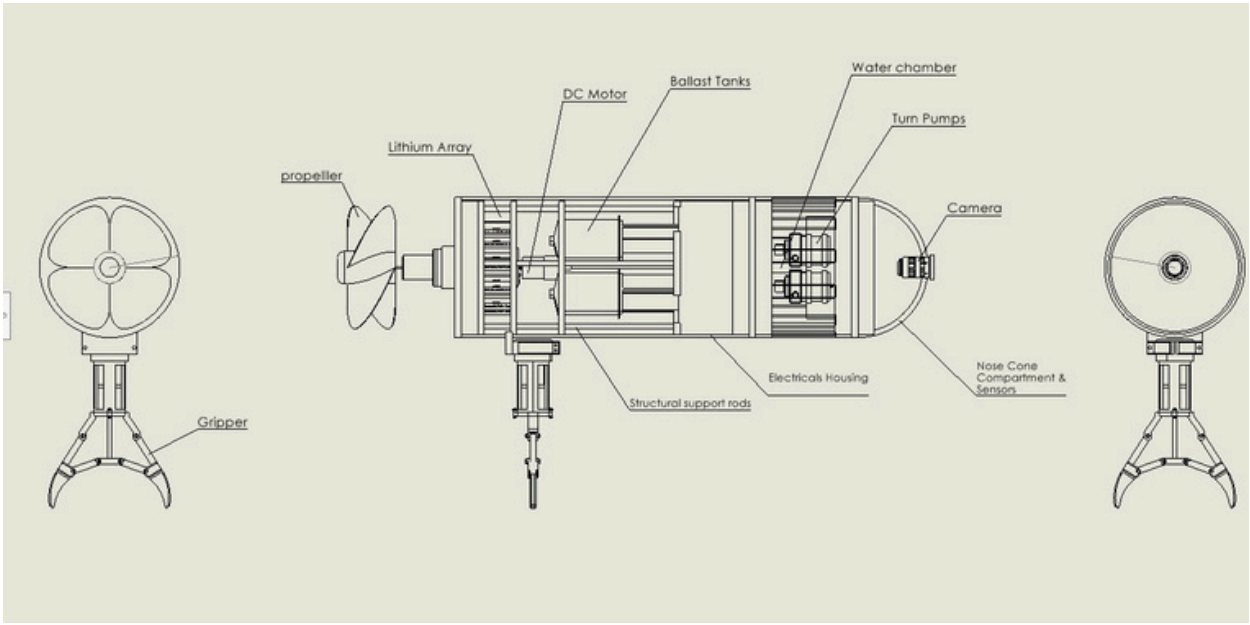
a. Length in cm - 60 cm b. Width in cm - 22cm c. Height in cm - 22cm

6. Timeline for Robot making with milestones. (Divided in activities vs. no. of days)

Activity	Duration (Days)	Milestones
1. Concept & Design	5	Finalize 3D CAD design, list all components, define control & navigation approach
2. Procurement & Logistics	7	Order electronics (Jetson/MCU, sensors, ESCs), thrusters, batteries, hull materials
3. Hull Fabrication & Waterproofing	10	Build main hull, nose, and rear sections; seal with O-rings/epoxy; test pressure tolerance
4. Power System Integration	5	Assemble battery, PDB, DC-DC converters; wire ESCs; test kill relay & e-stop
5. Computing & Control Setup	6	Configure Jetson/RPi & STM32; install OS, ROS, drivers; set up communication link
6. Sensor Integration	7	Connect camera, optical flow, IMU, pressure sensor; calibrate depth & orientation sensors
7. Thruster & Actuation Integration	6	Mount thrusters, piston ballast, and side fans; tune ESCs; verify PWM response
8. Software Development	12	Implement low-level motor drivers, PID control, and navigation logic (gate crossing, U-turn, ball drop)
9. Simulation & Testing (Gazebo/Pool)	8	Perform underwater simulations; shallow pool tests for buoyancy, movement, and stability
10. Debugging & Optimization	7	Fix leaks, power noise, and control errors; tune PID gains
11. Final Integration & Autonomous Mission Test	10	Full mission trial: gate pass → U-turn → ball drop; safety & recovery validation

★ **Total Estimated Duration: ~83 Days (~3 months)**

7. Please attach the proposedoutline (photograph) for understanding of the evaluation committee.



Link of drive with designs

https://drive.google.com/drive/folders/14DegDLPp76o3bq_kgattFHS9DsZbnBOk?usp=sharing

Ahmedabad University

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Ahmedabad University

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